

Art Simulates Life

3D Visualization Takes Pediatric Hospitalist Simulations to the Next Level

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ABSTRACT

Children's National Hospital is a leading pediatric teaching hospital with medical students, residency programs, fellowships, and research initiatives. We develop virtual 3D patients to improve medical training by simulating plausible, life-threatening scenarios in infants and children for Pediatric Hospitalists. Medical simulation is widely used to enhance the readiness of medical professionals [Davila and Price 2023; Motola et al. 2013]. Children's National's Division of Pediatric Hospital Medicine's Resuscitation Simulation Team (HRST) leads training for attending Hospitalists responsible for the inpatient care of sick and injured children across our entire regional network. Our 3D patient simulations provide realistic depictions of nuanced physical findings that are essential to identify uncommon, yet life threatening medical conditions in children. 3D models applicable for pediatric simulations are underrepresented compared to those used for adult patients.

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1 INTRODUCTION

Our first medical simulation project began during the early COVID-19 pandemic when a pivot to remote training was required. The simulation team had been working with manikins that offered hands-on training but could not effectively represent many medical conditions. While high fidelity simulation manikins allow for physical assessment and hands-on skills practice, they are limited in their presentation and require significant suspension of disbelief. Due to the ethical, emotional, and privacy-related challenges in capturing videos of children during active medical crises, realistic animated models were needed. Therefore, 3D depictions of critically



Figure 1: Infant with Congenital Hypothyroidism (left) Teen Suffers Overdose (right)

ill children with diverse ethnic backgrounds and highly specific medical conditions were created. This required a high level of skill to precisely mimic the physical findings, including skin changes, respiratory distress, tremors, and other abnormal movements to generate a comprehensive realistic appearance of each simulated patient. All were designed to prompt urgent action on the part of the participants. What was born out of necessity is now integral to the educational efforts of the HRST, which provides two unique 3D animated simulation sessions every year, averaging 50 hospitalists each quarter, promoting deeper diagnostic thinking in the context of a resuscitation.

2 STUDY POPULATION

The target audience are attending Pediatric Hospitalists responsible for patient care in the hospital setting, but the course could benefit any medical provider caring for sick children. This course is part of a curriculum that spans several years.

3 DESIGNS AND METHODS

3.1 Study Design

During instructor-led virtual patient simulation sessions at the hospital, 3D babies and child-sized patients are utilized. These virtual patients represent rare conditions that pediatric physicians may not have encountered previously, with interactive exams and feedback imitating the pressured situation of an acute medical crisis. Custom models and animation are developed in Autodesk Maya, Mudbox, Adobe After Effects and Photoshop, and timed to supporting breathing abnormalities, cries, and other sounds. Work created includes animation mimicking various respiratory diseases and

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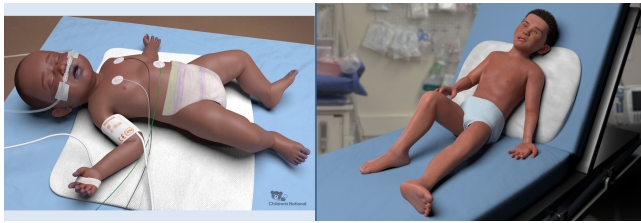


Figure 2: Impending Respiratory Failure (left) Boy Suffering Stroke (right)

other conditions, including animation of the patients, still images, hospital environments when appropriate, and interactive scenario cases built in Articulate Storyline. The 3D simulations are vetted by our physician specialists to ensure accuracy of the simulation. This is essential to the process due to the scarcity of videos available that accurately represent these clinical states in children.

3.2 Model Mechanics

Due to the lack of realistic pediatric models readily available for medical animation, three types of 3D models were developed: newborn, infant/baby, and child/adolescent. Pediatric Hospitalists provided granular details on medical signs and symptoms to the animator who added specifics into the models. We built details for inside of the mouths, expressive eyes and faces, incorporated unusual and distinct respiratory noises, and animated a myriad of medical problems. We created textures to mimic veins, bruising, discoloration, rashes, and eye changes.

The patient models in this course were animated using multiple custom-made blend shapes for the facial expressions, inner mouth and tongue movements, torso changes, and an underlying skeletal joint structure. The infant/baby model has overall body thickness blend shapes, making it possible to represent a young baby to toddler aged child. Set-driven expressions connect geometry deformations to the diaper or other clothing. A variety of skin textures were created depending on the ethnicities being portrayed and medical conditions. Animation was hand-keyed for all parts of the models and specifically timed to edited soundtracks of crying, breathing, gasping, or whatever the cases required. In addition, medical equipment such as ECG leads, blood pressure cuffs, and oxygen tubing were created and rigged to work with the body mechanics via constraints and set driven keys.

3.3 Animation and Graphics Applied in Case Studies

Storyline files are built to utilize graphics and animations. The facilitator guides the students through the course virtually via a live online platform. A brief patient history is given, followed by animation(s) and/or graphics of the patient. Then there is an interactive “exam” whereby students ask the facilitator to “examine” different areas of the patient image to reveal detailed findings, depending on the medical condition. A score sheet with areas “examined” and neglected are displayed after completion of the interactive exam. Feedback tallies are color-coded according to American Heart Association’s Pediatric Advanced Life Support Manual [American Heart

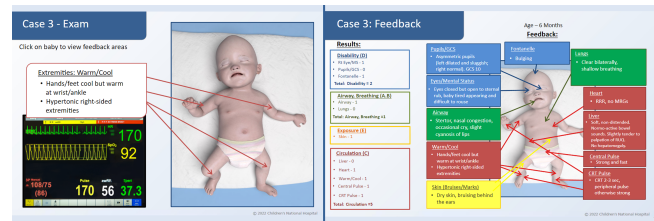


Figure 3: Examples of Case Exam (left) and Exam Feedback (right)

Association 2020]. After the initial exam and urgent interventions, participants are guided through the rest of the case, eventually arriving at the diagnosis and optimal treatment. Some cases simulate the condition changing over time, with a new set of animations, images, and exams. The conclusion reveals the correct diagnosis, and ultimate patient outcome.

4 RESULTS AND NEXT STEPS

In the past two and a half years, we have created six unique simulated stand-alone sessions with patient animation and/or graphics. Participants have uniformly reported that these realistic animations and images enhanced their learning and added a sense of urgency.

On a Likert scale, the vast majority strongly agreed that the cases were intellectually stimulating, and over 70% improved their comfort and confidence in managing various challenging medical situations. In at least one case, a Pediatric Hospitalist directly used our animations to aid in immediate visual recognition and treatment of a real-time emergency. Recently, these animated sessions were expanded to include nurses along with physician participants. Overall, we had 300 session participants and our curricula provided over 750 hours of continuing medical education which is required for medical re-licensure.

There are multiple plans to expand the impact of this course. The first plan is to make each animated simulation a stand-alone, self-guided curriculum that does not require a real-time facilitator, thus widely increasing course availability. To reach an international audience, we intend to publish several sessions on peer-reviewed online platforms.

Although this project began with a requirement for remote learning, due to the success of this course, we plan to continue developing virtual pediatric patients well into the future.

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